

Supplemental Information for

Geometry-mediated particle accumulation driven by nonhydrodynamic

viscosity effect with flow control implications in porous media

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I. DISCUSSIONS ON MODEL PARAMETERS

A. Rheology model

In the main manuscript, we systematically compare the accumulation behaviors of hard sphere and microgel particle suspensions applying their distinctly different rheological properties. For hard sphere suspensions, Krieger and Dougherty's model [1] with an exponent of $-2.5\phi_{\text{lim}}$ has been adopted. Fig. S1(a-b) presents differences among commonly used rheology models [2-4]. At higher volume fractions, models of Eilers and Maron-Pierce perform better. In the dilute regime considered in this work, predictions from Batchelor and Green, Krieger and Dougherty, and Eilers are consistent, whereas Maron-Pierce deviates. Moreover, since the rheology effect is too weak to trigger accumulation, the choice of the specific rheology model has a negligible impact on our discussions regarding accumulation states for hard sphere suspensions, as verified by our simulations under the same conditions (Fig. S1(c)).

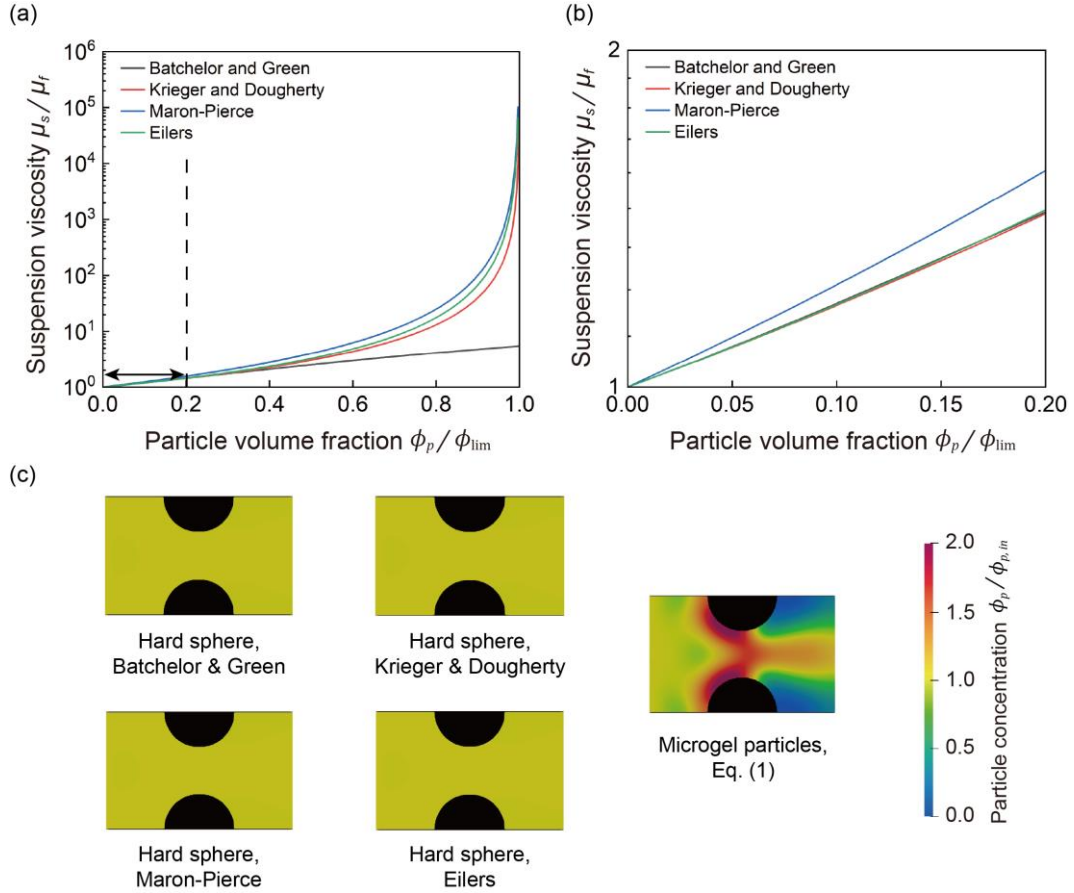


FIG. S1. Comparison of different rheology models of Batchelor and Green, of Krieger and Dougherty with exponent $-2.5\phi_{\text{lim}}$ and -2 (Maron-Pierce), and of Eilers for hard sphere suspensions. (a) Suspension

viscosity variations with respect to concentration. The dashed line and arrow denote the dilute condition considered in this work. (b) Detailed view of viscosity variations in the dilute regime. (c) Uniform concentration distribution applying all the above models in comparison to the strong accumulation by microgel particle suspension. Case setup: $U_{in} = 10^{-4}$ m/s, $d_p = 3$ μm , $\phi_{p,in} = 3.0$ vol%, $\chi = 0.4$.

It is interesting to note that if we consider high-concentration flow of hard sphere suspensions, geometry-mediated accumulation can also occur as viscosity becomes much more sensitive to particle concentration than in dilute conditions (Fig. S2), although the accumulation intensity is still weaker than the effects from microgel particles. However, the coupling effect of geometry-mediated accumulation, shear-induced migration, and clogging require further evaluation in this regime.

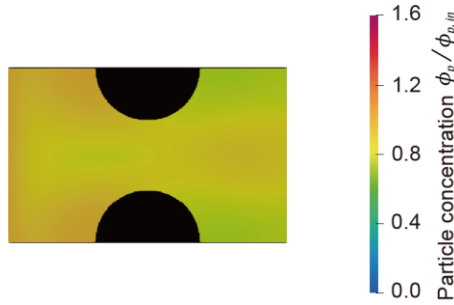


FIG. S2. Tests on the potential geometry-mediated accumulation effect in high-concentration flows of hard sphere suspensions (Maron-Pierce correlation [2] is adopted). Case setup: $U_{in} = 10^{-4}$ m/s, $d_p = 3$ μm , $\phi_{p,in} = 30$ vol%, $\chi = 0.4$.

In theoretical calculations, the viscosity of the particle phase is required. Considering the state before accumulation occurs, comparing the sum of Eq. (1c) and Eq. (1d) with Eq. (2b) yields $\mu_s \approx \phi_p \mu_p + \phi_f \mu_f$. Therefore, $\mu_p = (\mu_s - \phi_f \mu_f) / \phi_p$ is substituted into Eq. (7) as an estimate of the characteristic particle phase viscosity. This weighted average expression is only accurate when the velocity difference is small. However, the deviation is acceptable when applied as an approximate input for the accumulation number A_g , as validated by a wide range of numerical tests. In numerical simulations, we can directly apply the apparent suspension viscosity to obtain the evolution of particle transport. For microgel particles, $\mu_s = \mu_s(\dot{\gamma})$ and the characteristic shear rate $\dot{\gamma}$ for calculating A_g is estimated as U_{in}/W_{pore} .

B. Drag force model

The drag force coefficient β is another important parameter in both theoretical calculations and numerical simulations. In the Wen-Yu model, C_d and $|\mathbf{u}_f - \mathbf{u}_p|$ both depend on the flow condition, which exhibit opposing trends as the characteristic speed increases. In theoretical calculations, we use the suspension velocity at the throat $U = U_{in}/\chi$ for the characteristic velocity, since the velocity difference cannot be analytically given. As shown in Fig. S3, the velocity or Reynolds number dependence of β is negligible under the varied conditions.

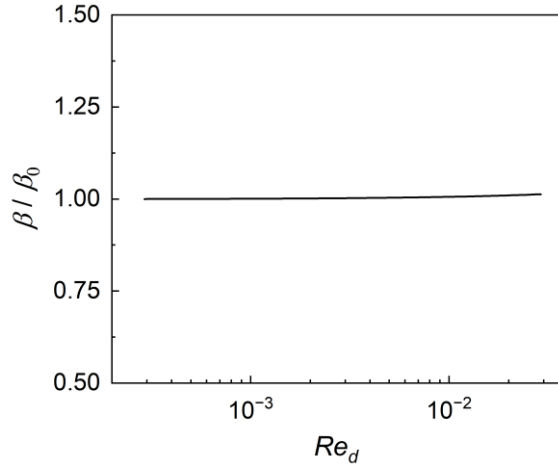


FIG. S3. Reynolds number dependence of the drag force coefficient. The parameters used are as follows: $U = 10^{-4} - 10^{-2}$ m/s, $\rho_f = 10^3$ kg/m³, $\phi_p = 3$ vol%, and $d_p = 3$ μ m for the calculation of $Re_d = \rho_f \phi_f d_p U / \mu_f$ and β . The velocity range is consistent with our simulation cases and β_0 corresponds to $U = 10^{-4}$ m/s.

This independence of drag force coefficient from the characteristic Reynolds number does not rely on the choice of specific drag force model. For the Batchelor model [5] also commonly used under Stokes flow condition and given by $\beta = 18\mu_f\phi_p(1 + 6.55\phi_p)/d_p^2$, the velocity independence is clearer.

As shown in Fig. S4, we further confirmed that the simulated relative drift velocities by the Wen-Yu model and the Batchelor model are quantitatively consistent under typical Stokes flow conditions, indicating the reliability and generality of our results.

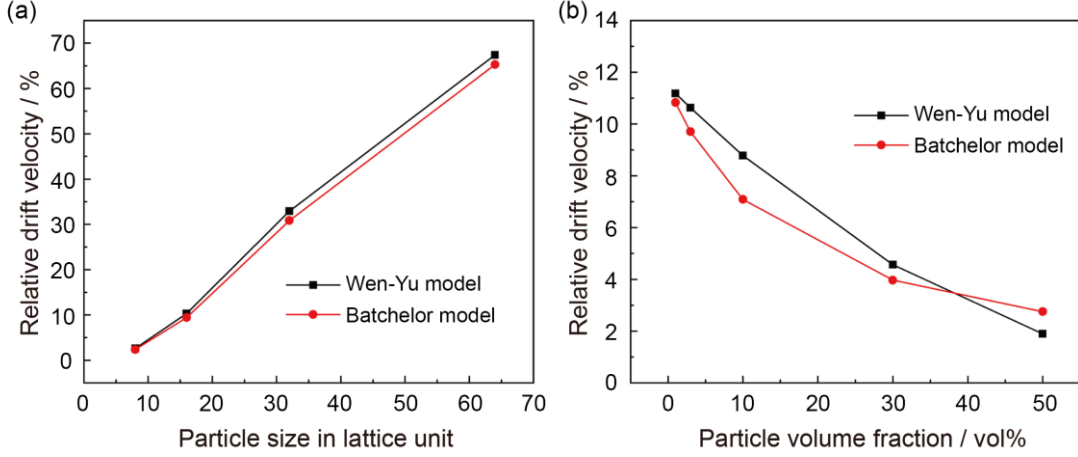


FIG. S4. Comparison of the simulated relative drift velocity (defined as the ratio of the average velocity difference between two phases to the velocity of the fluid phase) applying two different drag models. Simulations are performed under pressure-driven Poiseuille flow in a 2D straight channel with fixed particle phase viscosity. (a) Relative drift velocity versus particle size (expressed in lattice unit) when $\phi_p = 3$ vol%. (b) Relative drift velocity versus particle volume fraction when $d_p^* = 16$. The corresponding fluid Reynolds number $Re_f = \rho_f U_f W / \mu_f \ll 1$.

II. DETAILS OF NUMERICAL METHOD

As introduced in the main manuscript, we solve the governing equations for the fluid phase (Eq. (1a) and (1c)) and the suspension phase (Eq. (2a) and (2b)) based on the mixture-rheology two-fluid model (mrTFM). Information regarding the particle phase (ϕ_p and $\mathbf{u}_p$) can be calculated by local mass and momentum conservation, which serves as an input for the drag force calculation. The lattice Boltzmann method (LBM) is adopted with GPU computing to improve computational efficiency. All simulations are solved using a standard D2Q9 lattice set. More details about the algorithm implementation and benchmarks can be found in our previous work [6].

The lattice Boltzmann equation with multiple relaxation times (MRT) can be expressed as

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) - f_i(\mathbf{x}, t) = -\mathbf{M}^{-1} \mathbf{S}(\mathbf{m}(\mathbf{x}, t) - \mathbf{m}^{\text{eq}}(\mathbf{x}, t)) + \Delta t \mathbf{M}^{-1} \left(\mathbf{I} - \frac{\mathbf{S}}{2} \right) \mathbf{G}, \quad (\text{S1})$$

where $f_i(\mathbf{x}, t)$ is the density distribution function at the space-time position $(\mathbf{x}, t)$, moving along the i th direction by discrete velocities $\mathbf{e}_i$ (unit vector). $\mathbf{M}$ is the transformation matrix

to the moment space [7], $\mathbf{m} = \mathbf{M}\mathbf{f}$ is the mapped moment of distribution functions, and $\mathbf{m}^{\text{eq}} = \mathbf{M}\mathbf{f}^{\text{eq}}$. $\mathbf{G} = \mathbf{M}\hat{\mathbf{F}}$ accounts for the forcing term in the moment space, where $\hat{\mathbf{F}}$ is calculated using Guo's forcing scheme [8,9]. $\mathbf{I}$ is the unit tensor and $\mathbf{S}$ is a diagonal matrix composed of relaxation parameters [10].

The macroscopic variables can be related to f_i as

$$\rho = \sum f_i, \rho \mathbf{u} = \sum_i f_i \mathbf{e}_i + \frac{\Delta t}{2} \mathbf{F}. \quad (\text{S2})$$

The body force only needs to be introduced in the evolution equation of the fluid phase for the mrTFM, which includes two key terms

$$\mathbf{F}_f = P\nabla\phi_f - \beta(\mathbf{u}_f - \mathbf{u}_p), \quad (\text{S3})$$

where the first term is the correction for pressure gradient considering volume average [11] and the second term is the drag force. The gradient terms involved in the algorithm are calculated employing the isotropic gradient scheme with second-order approximation [12].

The microgel particle suspension is a typical non-Newtonian fluid with shear-dependent viscosity. The shear rate $\dot{\gamma}$ can be expressed as

$$\dot{\gamma} = (2|\Psi_{\alpha\beta}\Psi_{\alpha\beta}|)^{\frac{1}{2}}, \Psi_{\alpha\beta} = \frac{3}{2\rho\tau c^2} \sum_i f_i^{(1)} \mathbf{e}_{i\alpha} \mathbf{e}_{i\beta}, \quad (\text{S4})$$

where Ψ is the shear rate tensor, τ is the relaxation time, $c = \Delta x/\Delta t = 1$ is the lattice speed, and $f_i^{(1)} \approx f_i - f_i^{\text{eq}}$ [13].

We have experimentally validated the capability of our model to capture particle transport and distribution patterns, as summarized in Fig. S5. The fair consistency between simulations and microfluidic experiments demonstrates the applicability of our model for predicting microgel particle transport behaviors under realistic conditions. Fig. S5(a) presents the local view of particle suspension flow through a regular porous medium composed of uniform posts. Particle accumulation is observed at the throat entrance driven by fluid acceleration from pore to throat, which becomes significant at the edge region with a wavy concentration distribution along the flow direction. The agreement in particle concentration distributions along both the

flow and cross-stream directions shown in Fig. S5(b-c) is quantitatively satisfactory given the impurity of microgel particle suspensions in experiments and the complexity of the systems. More details of the experimental system setup can be found in our previous work [6].

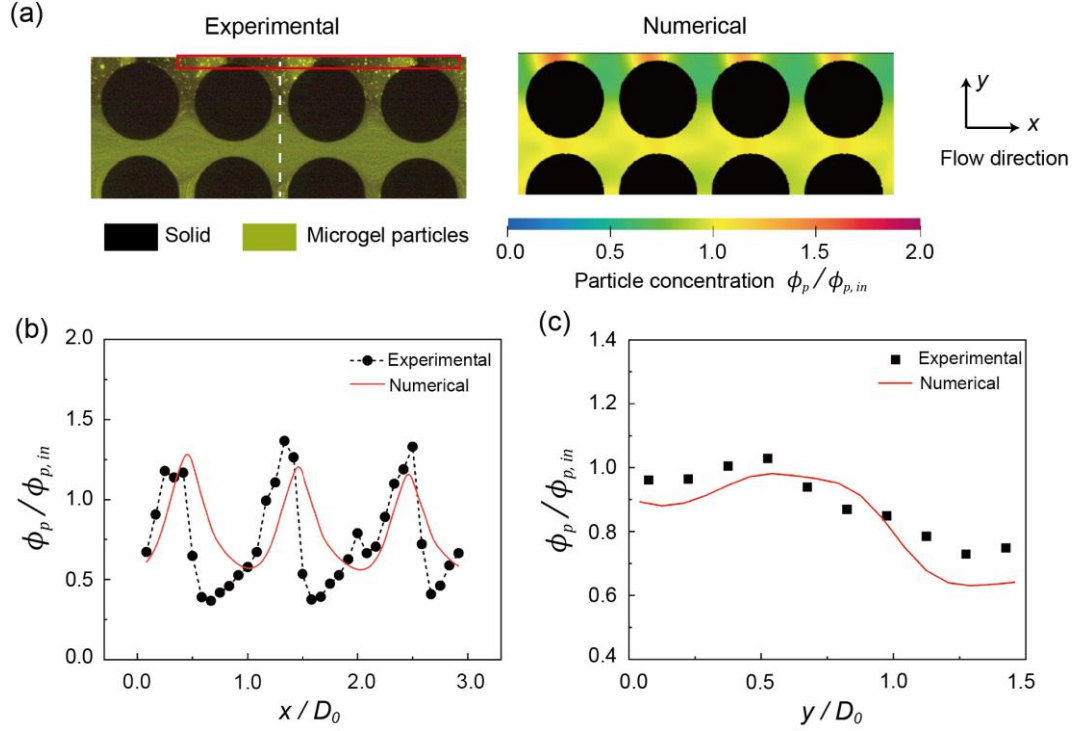


FIG. S5. Comparison of microgel particle concentration distributions from microfluidic experiments and simulations in the edge region of a regular porous medium during suspension flow. For experimental quantification, data points are obtained by locally averaging the grayscale values from images. A linear relationship between light intensity and concentration is applied to calculate local concentrations [14,15], with the value corresponding to injection concentration as a reference. The measured mean particle diameter in experiments is $3.3 \mu\text{m}$ with a distribution ranging from 1 to $10 \mu\text{m}$. The characteristic particle size is set to be $3 \mu\text{m}$ in simulations and the grid size is $6 \mu\text{m}$. The minimum throat size is $60 \mu\text{m}$ at the edge region. (a) Local images from experiments and simulation results. (b) Particle concentration distribution along the flow direction corresponding to the x position range denoted by red boxes in (a). Positions closest to the inlet are excluded from the comparison considering differences in inlet buffer settings between experiments and simulations. (c) Particle concentration distribution along the cross-stream direction corresponding to the y position range denoted by white dashed lines in (a). The concentration distributions are normalized by the injection concentration $\phi_{p,in} = 3.0 \text{ vol\%}$ in all cases, and D_0 is the center distance between cylinders.

III. SIMULATION AND CASE SETUP

A. Boundary conditions

The standard bounce-back scheme is applied to achieve non-slip condition at solid boundaries (including top and bottom wall). It is worth mentioning that wall slip effects have been widely reported in rheological characterizations of concentrated suspensions [16-18], owing to the formation of a depleted layer. Given the dilute injection conditions (≤ 5 vol%) considered in this work, we adopt the non-slip condition in all simulation cases, as adopted in previous literatures regarding particle transport in porous media [19-21].

The Zou-He boundary [22] is applied with constant velocity at the inlet and constant pressure at the outlet. The volume fraction of the particle phase is fixed at the inlet and the zero-gradient Neumann boundary is applied at the outlet to achieve free outflow condition [23]. For suspension flow, the entire region is filled with water initially and microgel particles are injected at a fixed concentration.

B. Case setup for pore-throat geometries

Table S1 presents the detailed parameter setup for numerical tests in typical pore-throat geometries. The parameter sets and the corresponding dimensionless numbers are listed for microgel particle and hard sphere suspensions, respectively. The accumulation number A_g is calculated from Eq. (7), and the Stokes number is $Stk = \rho_p U_{in} d_p^2 / (18\mu_f W_{pore})$. The grid size is $6 \mu\text{m}$ in all cases.

TABLE S1. Parameter sets and dimensionless numbers for cases of pore-throat geometries.

χ	U_{in} (m/s)	d_p (μm)	$\phi_{p,in}$ (vol%)	Stk	A_g (microgel particles)	A_g (hard spheres)
0.4	10^{-4}	6	3	1.67×10^{-7}	7.08×10^0	9.05×10^{-4}
0.4	10^{-3}	6	3	1.67×10^{-6}	1.15×10^0	8.99×10^{-4}
0.4	10^{-2}	6	3	1.67×10^{-5}	1.83×10^{-1}	8.71×10^{-4}
0.4	10^{-4}	6	1	1.67×10^{-7}	2.20×10^0	9.08×10^{-4}

0.4	10^{-3}	6	1	1.67×10^{-6}	3.58×10^{-1}	9.02×10^{-4}
0.4	10^{-2}	6	1	1.67×10^{-5}	5.70×10^{-2}	8.74×10^{-4}
0.4	10^{-4}	3	3	4.17×10^{-8}	1.77×10^0	2.26×10^{-4}
0.4	10^{-3}	3	3	4.17×10^{-7}	2.89×10^{-1}	2.26×10^{-4}
0.4	10^{-2}	3	3	4.17×10^{-6}	4.64×10^{-2}	2.21×10^{-4}
0.4	10^{-4}	3	1	4.17×10^{-8}	5.50×10^{-1}	2.27×10^{-4}
0.4	10^{-3}	3	1	4.17×10^{-7}	8.98×10^{-2}	2.26×10^{-4}
0.4	10^{-2}	3	1	4.17×10^{-6}	1.45×10^{-2}	2.22×10^{-4}
0.6	10^{-4}	6	3	1.67×10^{-7}	2.10×10^0	2.68×10^{-4}
0.6	10^{-3}	6	3	1.67×10^{-6}	3.42×10^{-1}	2.67×10^{-4}
0.6	10^{-2}	6	3	1.67×10^{-5}	5.47×10^{-2}	2.61×10^{-4}
0.6	10^{-4}	6	1	1.67×10^{-7}	6.52×10^{-1}	2.69×10^{-4}
0.6	10^{-3}	6	1	1.67×10^{-6}	1.06×10^{-1}	2.68×10^{-4}
0.6	10^{-2}	6	1	1.67×10^{-5}	1.70×10^{-2}	2.61×10^{-4}
0.6	10^{-4}	3	3	4.17×10^{-8}	5.25×10^{-1}	6.71×10^{-5}
0.6	10^{-3}	3	3	4.17×10^{-7}	8.56×10^{-2}	6.69×10^{-5}
0.6	10^{-2}	3	3	4.17×10^{-6}	1.38×10^{-2}	6.59×10^{-5}
0.6	10^{-4}	3	1	4.17×10^{-8}	1.63×10^{-1}	6.73×10^{-5}
0.6	10^{-3}	3	1	4.17×10^{-7}	2.66×10^{-2}	6.71×10^{-5}
0.6	10^{-2}	3	1	4.17×10^{-6}	4.31×10^{-3}	6.61×10^{-5}
0.8	10^{-4}	6	3	1.67×10^{-7}	8.85×10^{-1}	1.13×10^{-4}
0.8	10^{-3}	6	3	1.67×10^{-6}	1.44×10^{-1}	1.13×10^{-4}
0.8	10^{-2}	6	3	1.67×10^{-5}	2.32×10^{-2}	1.11×10^{-4}
0.8	10^{-4}	6	1	1.67×10^{-7}	2.75×10^{-1}	1.14×10^{-4}
0.8	10^{-3}	6	1	1.67×10^{-6}	4.49×10^{-2}	1.13×10^{-4}
0.8	10^{-2}	6	1	1.67×10^{-5}	7.23×10^{-3}	1.11×10^{-4}
0.8	10^{-4}	3	3	4.17×10^{-8}	2.21×10^{-1}	2.83×10^{-5}
0.8	10^{-3}	3	3	4.17×10^{-7}	3.62×10^{-2}	2.82×10^{-5}
0.8	10^{-2}	3	3	4.17×10^{-6}	5.85×10^{-3}	2.79×10^{-5}

0.8	10^{-4}	3	1	4.17×10^{-8}	6.88×10^{-2}	2.84×10^{-5}
0.8	10^{-3}	3	1	4.17×10^{-7}	1.12×10^{-2}	2.83×10^{-5}
0.8	10^{-2}	3	1	4.17×10^{-6}	1.82×10^{-3}	2.80×10^{-5}

C. Validation of grid resolution used for simulations in porous media

The computational domain of the regular dual-permeability model is resolved by 1620×960 grids. To ensure grid independence, we further conduct simulations at a series of grid resolutions. The real physical properties and boundary conditions of the system remain unchanged as the grid resolution varies. The simulated particle distributions at a specific time are systematically compared. As shown in Fig. S6, the simulation results become inaccurate with unreasonable particle accumulation when the grid resolution is too low. However, the concentration distribution and standard deviation remain stable as the grid resolution increases beyond 1620×960 , confirming the reliability of results presented in this work.

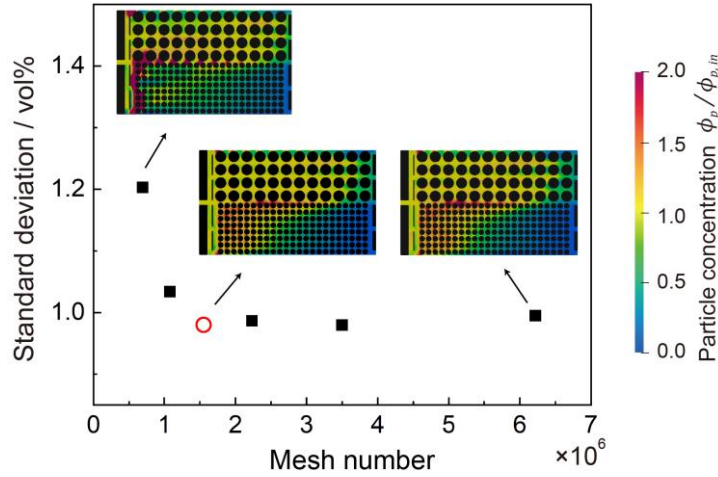


FIG. S6. Grid independence test. The standard deviations of particle concentration are compared to reflect differences in distribution patterns. The inset images show the global concentration distribution corresponding to different grid resolutions (denoted by the arrows). The red open circle denotes the grid resolution applied in our simulations. The results are obtained after 0.9 PV of injection when $d_p = 3 \mu\text{m}$ and $\phi_{p,in} = 3.0 \text{ vol\%}$.

IV. DISCUSSIONS ON SIMULATION RESULTS

In our theoretical analysis of the accumulation number, we show that for pronounced nonhydrodynamic viscosity effects as in microgel particle suspensions, the dominant driving force for the particle phase should be the interphase drag rather than the pressure gradient. We have validated this argument using statistics from simulation results. Note that in simulations we solve the suspension and fluid phases, whereas in theoretical analysis we focus on the momentum balance of the particle phase. As introduced in the main manuscript, information regarding the particle phase (ϕ_p and $\mathbf{u}_p$) can be directly obtained from local mass and momentum conservations. As shown in Fig. S7, for a typical case of pore-throat geometries exhibiting strong accumulation ($U_{in} = 10^{-4}$ m/s, $d_p = 3$ μm , $\phi_{p,in} = 1.0$ vol%, $\chi = 0.4$), the drag force term is indeed much larger than the pressure gradient term for the particle phase, with both calculated from the output flow field data.

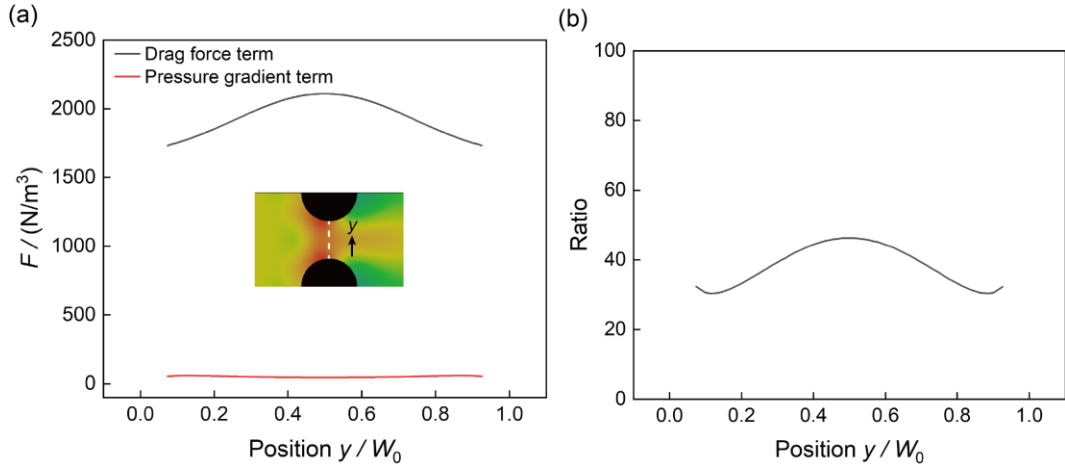
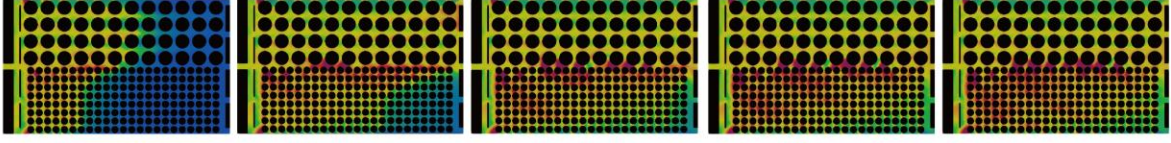


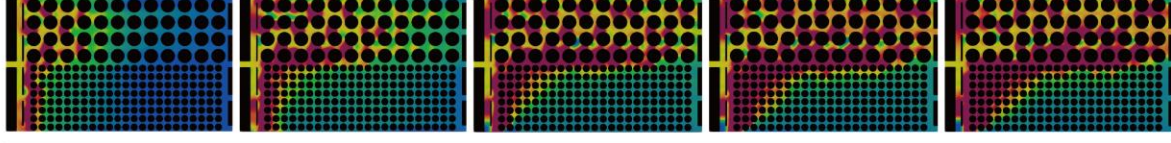
FIG. S7. Statistics of the drag force and the pressure gradient terms for the particle phase showing the dominant driving force: (a) absolute values of the two terms and (b) the relative ratio of drag force to pressure gradient. All terms are calculated based on Eq. (9) from simulation data.

Fig. S8 presents the time evolution of concentration distribution during injection. Differences in accumulation patterns in Regime II and Regime III are not influenced by the injection time. As shown in Fig. S8(b), the localized accumulation in the LPL gradually extends towards downstream but at a very slow rate, indicating the distinction between the filter-cake-like distribution induced by accumulation and real filter cakes formed by physical clogging.

(a) $d_p = 3 \mu\text{m}$, $\phi_{p,in} = 5.0 \text{ vol\%}$ (Regime II)



(b) $d_p = 6 \mu\text{m}$, $\phi_{p,in} = 5.0 \text{ vol\%}$ (Regime III)



0.6 PV 1.2 PV 1.8 PV 2.4 PV 3.0 PV
Injection time

FIG. S8. Time evolution of concentration distribution under different accumulation states.

To evaluate the impact of disorder on accumulation behaviors, we conduct a series of simulations in a disordered medium generated by the QSGS method [24]. As shown in Fig. S9, size and concentration effects on accumulation states are consistent with observations in regular media. As particle size and injection concentration increase, particle accumulation becomes stronger and filter-cake-like intensified accumulation gradually forms in the upstream region.

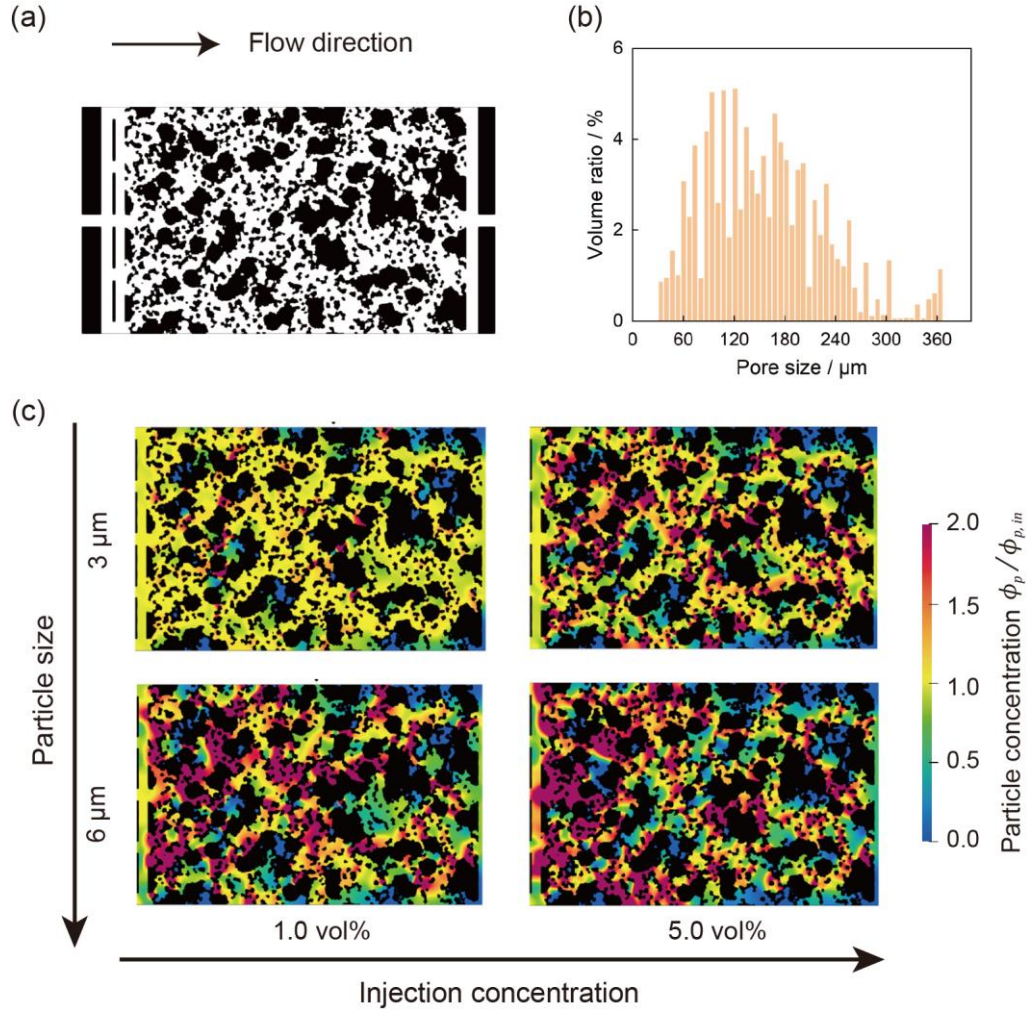


FIG. S9. Example cases of microgel particle accumulation in disordered media. (a) A randomly generated disordered medium with complex geometry. The grid resolution is set the same as in the dual-permeability media. (b) Pore size distribution characterized by the maximal ball method. (c) Accumulation patterns after 1.8 PV of injection at typical particle sizes and injection concentrations.

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